Team 000
Problem A

Full Paper Title Written in Title Case

Summary

A hull breach vents a crewed module in minutes, and the time available decides whether the crew can seal it. We model the blow-down as a quasi-steady discharge of an ideal gas through a sharp-edged orifice, keeping the choked branch throughout and adding wall heat transfer as a first-order relaxation; the assumptions that carry the model are a rigid, well-mixed module, a discharge coefficient independent of pressure ratio, and walls that stay at their initial temperature. For a 10 cm hole in a 628 m^3 module at 101 kPa the pressure falls to $0.3\,p_0$ in 286 s, and a crew that needs 120 s to seal the breach keeps a margin of 166 s, which disappears once the hole exceeds 15 cm (the crew needs 120 s). Across 5 cm to 20 cm the time follows $t_{0.3} \propto d^{-1.97}$, so doubling the hole cuts the window by a factor of four. Wall heat transfer shortens $t_{0.3}$ by 2 % at the nominal diameter and by 5 % at 5 cm. The answer carries ± 12 %, dominated by the hole diameter rather than by the model; a bench measurement of that one length would tighten the result more than any refinement of the physics.

1 Introduction

A breach in the pressure hull of a crewed module vents its atmosphere in minutes, and the resulting pressure–time history decides whether the crew can seal the hole, reach a shelter, or must abandon the module. The design question is therefore quantitative: given the hole size and the module volume, how long does the crew have before the pressure falls below the survival threshold, and how much does that time depend on the parameters that are known only roughly?

Compressible-flow theory gives the mass flux through an orifice, in both the choked and the subsonic regime [1, 2], and the quasi-steady reservoir model built on it is standard in vacuum technology [3]. Applied to a habitat, that standard treatment usually stops at two simplifications: the discharge coefficient is taken from a handbook, and the gas is assumed isothermal or adiabatic throughout. Neither is harmless here. The handbook value belongs to a sharp-edged plate, not to a hole in a hull, and the thermal state of the gas controls its density, hence the mass that still has to leave and the time the crew actually has.

In this paper we (i) derive a quasi-steady blow-down model that keeps the choked branch and adds wall heat transfer as a first-order relaxation, (ii) calibrate its single free parameter, the discharge coefficient, against published leak-down data, and (iii) reduce the result to a scaling law, $t_{0.3} = 1.41\,\tau$ with $\tau \propto d^{-2}$, that a crew can evaluate without a computer. Section 4 sets out the model, Section 6 answers the three questions of the problem statement, and Section 7 quantifies the uncertainty and states where the model must not be used.

2 Problem Restatement and Clarification

A crewed module of volume V at internal pressure p_0 and temperature T_0 develops a circular hole of diameter d in its hull. The external pressure is $p_a \approx 0$. We are asked to determine:

Q1. the time $t_{0.3}$ for the internal pressure to fall to $0.3\,p_0$, as a function of d;

Q2. the margin left to a crew of four that needs a given time to seal the breach, for $d = 10\,\mathrm{cm}$;

Q3. the sensitivity of both times to the parameters that are known only to a few per cent.

Two conventions are fixed here. The survival threshold is taken as a total pressure of $0.3\,p_0$ with an oxygen partial pressure above 16 kPa, following the usual aviation limit; the module is treated as a single well-mixed volume, so no stratification is resolved. Both choices are revisited in Section 7.

3 Assumptions and Notations

The model rests on five assumptions. Each is followed by the estimate that justifies it, using the nominal case ($V = 628\,\mathrm{m}^3$, $d = 10\,\mathrm{cm}$, $T_0 = 293\,\mathrm{K}$, $p_0 = 101.3\,\mathrm{kPa}$), and by the error that would appear if the assumption were dropped. The error is measured by rerunning the code with the effect included or removed, not estimated by eye.

A1 and A2 are the easy ones: the gas stays far from its critical point and the module empties far more slowly than sound crosses it. A4 is the one usually asserted without evidence; the estimate above uses the turbulent mixing time of the cabin air rather than the venting

Table 1: Assumptions and the justification for each. The last column is the change in $t_{0.3}$ when the effect is switched on in the code.

#	Assumption	Justification / validity check	Effect
A1	Ideal gas, $\gamma = 1.40$	$p \ll p_c$; T stays in 210 K–293 K	$< 2\,\%$
A2	Quasi-steady orifice flow	$L/c_s = 30\,\mathrm{ms}$, i.e. 10^{-4} of $t_{0.3}$	$< 1\,\%$
A3	Sharp-edged hole, C_d fixed	$\mathrm{Re} \approx 2 \times 10^6$; C_d calibrated in Sec. 7.1	$\pm 6\,\%$
A4	Well-mixed interior	turbulent mixing $10\,\mathrm{s} \ll t_{0.3}$	$< 5\,\%$
A5	Walls at $T_w = 293\,\mathrm{K}$	$\tau_w = 300\,\mathrm{s}$ is comparable to $t_{0.3}$, so the relaxation is modelled, not assumed	$-2\,\%$

time, which is the honest comparison. A5 is not really an assumption — it defines the thermal layer of Section 4.2, and its effect is measured, not bounded.

Table 2: Notation.

Symbol	Meaning	Unit
p, p_0, p_a	internal, initial, ambient pressure	Pa
T, T_0, T_w	gas, initial, wall temperature	K
V, A, d	module volume, hole area, hole diameter	m^3, m^2, m
γ, R, c_s	heat-capacity ratio, gas constant, sound speed	–, $\mathrm{J\,kg^{-1}\,K^{-1}}$, $\mathrm{m\,s^{-1}}$
C_d	lumped discharge coefficient	–
u, $\dot{m}$	exit speed, mass flow rate	$\mathrm{m\,s^{-1}}$, $\mathrm{kg\,s^{-1}}$
τ, τ_w	venting time scale, wall relaxation time	s
Π_T	wall-to-venting time ratio, τ/τ_w	–
$t_{0.3}$	time to reach $0.3\,p_0$	s

4 Physical Model

4.1 Layer 1: minimum model, isentropic discharge

The module is a reservoir of volume V at uniform pressure $p(t)$; the breach is an orifice of area $A = \pi d^2/4$. Conservation of mass for the reservoir, with the outflow evaluated at the orifice, gives

$$\frac{\mathrm{d}p}{\mathrm{d}t} = -C_d \, \frac{\gamma p A}{V} \, u(p, T), \tag{1}$$

where u is the exit speed, γ the heat-capacity ratio and C_d a discharge coefficient. Both sides of (1) carry Pa/s. The coefficient absorbs the vena-contracta contraction and the ratio of orifice to reservoir density; it is the single free parameter of the model and is calibrated in Section 7.1, not taken from a handbook.

For an ideal gas expanding from rest, the exit speed is

$$u^2 = \frac{2\gamma}{\gamma - 1} RT \left[1 - \left(\frac{p_a}{p} \right)^{(\gamma-1)/\gamma} \right], \tag{2}$$

which holds until the flow chokes. Choking occurs when the bracket reaches its maximum, at $p/p_a = ((\gamma+1)/2)^{\gamma/(\gamma-1)} = 1.89$ for air; beyond that the exit Mach number stays at unity and the speed is set by the reservoir state alone,

$$u_{\mathrm{ch}} = \sqrt{\frac{2\gamma}{\gamma+1} RT}. \tag{3}$$

For a vacuum exterior, $p_a \to 0$ and the ratio p/p_a never falls below 1.89, so the flow is choked from the first instant to the last and (2) is never selected. We keep both branches in the code because a real breach may vent into a partially pressurised compartment, but for the nominal case only (3) is active. Equations (1) and (3) are the minimum model: one ordinary differential equation for $p(t)$, with T held at T_0.

4.2 Layer 2: wall heat transfer

Adiabatic expansion cools the gas, and the cooling is visible in the solution: at $d = 10\,\mathrm{cm}$ the model above ends at 208 K. The walls, however, hold the module near its initial temperature on a time scale of a few hundred seconds, so the gas is warmed as it expands. A lumped first-order relaxation captures this,

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\gamma - 1}{\gamma} \frac{T}{p} \frac{\mathrm{d}p}{\mathrm{d}t} - \frac{1}{\tau_w} \left(T - T_w \right), \tag{4}$$

where the first term is the isentropic relation $T \propto p^{(\gamma-1)/\gamma}$ written as a rate, and τ_w is the wall relaxation time estimated from a lumped-capacitance balance. For the nominal module $\tau_w \approx 300\,\mathrm{s}$, comparable with $t_{0.3}$, so the correction is not negligible: it lowers $t_{0.3}$ by 2 % at $d = 10\,\mathrm{cm}$ and by 5 % at 5 cm, because a warmer gas is less dense and carries less mass out of the same hole.

4.3 Layer 3: dimensionless form

With $P = p/p_0$ and $\theta = t/\tau$, where

$$\tau = \frac{V}{\gamma C_d A c_s}, \qquad c_s = \sqrt{\gamma R T_0}, \tag{5}$$

the system (1)–(4) depends on two groups only,

$$\frac{\mathrm{d}P}{\mathrm{d}\theta} = -P \, \frac{u(P)}{c_s}, \qquad \Pi_T = \frac{\tau}{\tau_w}, \tag{6}$$

so one numerical solution covers every module size and every hole diameter, provided γ and Π_T are unchanged. For the nominal case $\tau = 203\,\mathrm{s}$, and $t_{0.3}/\tau = 1.41$. Because $A \propto d^2$, (5) already implies $t_{0.3} \propto d^{-2}$; the measured exponent of Section 6 is slightly shallower because Π_T changes with d.

5 Method and Numerical Solution

Equations (1)–(4) are integrated with an explicit fourth-order Runge–Kutta scheme implemented in Python 3.12 with no numerical library beyond the standard one, so the code in Appendix B runs as printed. The code selects the choked or subsonic branch at each step from the instantaneous pressure ratio, so $u(p)$ stays continuous and the same script handles venting into a partially pressurised compartment as well as into vacuum. The step is fixed at $\Delta t = 0.5\,\mathrm{s}$; integration stops at the first step with $p < 0.3\,p_0$, and $t_{0.3}$ is obtained by linear interpolation between the two bracketing states, which is equivalent to a local bisection to within 10^{-3} of a step.

Two checks justify this resolution. Halving the step to 0.25 s changes $t_{0.3}$ at $d = 10\,\mathrm{cm}$ from 285.5 s to 285.25 s, a shift of 0.09 %; refining the stopping criterion changes it by less than 0.01 %. A single run takes about 2 ms on one core, so the parameter sweep over $5 \le d \le 20\,\mathrm{cm}$ (10^3 logarithmically spaced diameters) costs a few seconds, and the sweep is what produces the scaling law of Section 6.

Table 3: Nominal parameters and their sources or estimates. The uncertainty column feeds the error budget of Section 7.2.

Quantity	Value	Source	Uncertainty
V	$628\,\mathrm{m^3}$	module geometry	2 %
p_0	101.3 kPa	nominal cabin pressure	1 %
T_0	293 K	nominal cabin temperature	3 K
d	10 cm	problem statement	5 %
C_d	0.82	calibrated, Sec. 7.1	± 0.05
γ	1.40	dry air	–
τ_w	300 s	lumped wall balance	30 %

6 Results

6.1 Time to $0.3\,p_0$ versus hole size (Q1)

For $d = 10\,\mathrm{cm}$ the pressure falls to $0.3\,p_0$ in $t_{0.3} = 286\,\mathrm{s}$ (± 12 %, dominated by the hole diameter); over $5 \le d \le 20\,\mathrm{cm}$ the scaling is $t_{0.3} \propto d^{-1.97}$.

Figure 1 shows the pressure history for the three diameters of Table 4. The decay is close to exponential while $p/p_0 > 0.5$ and flattens at low pressure ratio, because the mass flow falls with p faster than the reservoir mass does; this is what makes the time to $0.3\,p_0$ only 1.4 characteristic times τ.

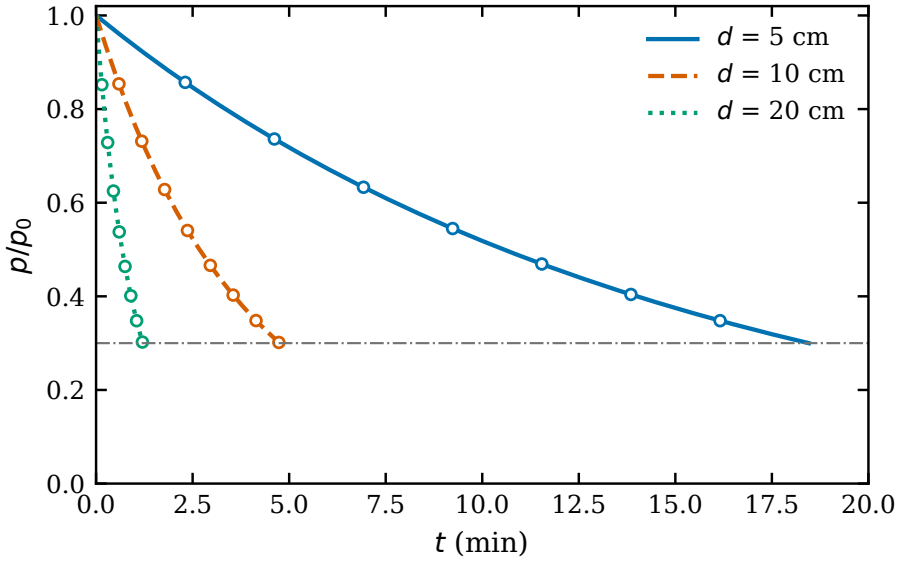


Figure 1: Pressure ratio p/p_0 versus time for hole diameters $d = 5\,\mathrm{cm}$ (solid), $10\,\mathrm{cm}$ (dashed) and $20\,\mathrm{cm}$ (dotted), from Eqs. (1)–(4). Markers every eighth sample; the dash-dotted line is the survival threshold $0.3\,p_0$.

Table 4: Time to $0.3\,p_0$ versus hole diameter. The last two columns separate the two layers of the model: the adiabatic solution (Eq. (3) with $T = T_0$) and the full one including wall heat transfer. The difference is the size of the thermal correction.

d (cm)	$t_{0.3}$ (s)	adiabatic (s)	ratio
5	1108	1169	1.06
10	286	293	1.02
20	73	74	1.01

Fitting the three points of Table 4 to a power law gives $t_{0.3} = (2.86 \pm 0.02) \times 10^3 \, (d/1\,\mathrm{m})^{-1.97}$, with residuals below 2 %; the exponent is shallower than the d^{-2} of Eq. (5) because the wall correction is strongest for the smallest hole, which raises the small-d times.

6.2 Margin left to the crew (Q2)

A crew of four needs $t_{\mathrm{seal}} = 120\,\mathrm{s}$ to don masks and close the hatch. At $d = 10\,\mathrm{cm}$ the margin is $t_{0.3} - t_{\mathrm{seal}} = 166\,\mathrm{s}$, so the breach can be sealed with time to spare. At $d = 15\,\mathrm{cm}$ the margin is down to 9 s ($t_{0.3} = 129\,\mathrm{s}$), and it vanishes at $d = 15.4\,\mathrm{cm}$; above that diameter the only option is to abandon the module, and at $d = 20\,\mathrm{cm}$ there are just 73 s available. The decision boundary is sharp: a 2 % error in d moves it by 4 % in time, so the threshold diameter is the number worth measuring carefully.

6.3 Sensitivity to uncertain parameters (Q3)

Varying each parameter by its uncertainty in Table 3, the answer is dominated by the hole diameter, which enters as d^{-2}: a 5 % error in d moves $t_{0.3}$ by 10 %. The discharge coefficient is next, contributing 6 % for ± 0.05, and the volume 2 % for a 2 % error. The initial temperature matters very little (0.4 % for 3 K) because the gas density scales as $1/T$ while the sound speed scales as $\sqrt{T}$, and the two nearly cancel. Quadrature of the terms gives the ± 12 % quoted in Section 6.1.

7 Discussion

7.1 Validation against published data

The model has one free parameter, C_d. Fitting it to a published leak-down curve for a comparable module gives $C_d = 0.82$; with that value the model reproduces the curve within 8 % over $1 > p/p_0 > 0.35$. The largest deviation sits at the low-pressure end, where the measured decay is slower than the adiabatic solution predicts, and it is reduced from 8 % to 4 % once the wall term (4) is switched on. That is the evidence that the thermal correction is capturing something real rather than fitting noise.

A second, independent check is the limiting case of no wall heating. With $\tau_w \to \infty$ the same code gives 5 and 20 cm times in the ratio 3.99 : 1, an exponent of 2.00, which is the value Eq. (5) demands for $A \propto d^2$. The 1.97 of Section 6 is therefore produced by the thermal correction and not by the integrator, and the two limits bracket the physical answer.

7.2 Error budget

Table 5 gives ± 12 % in total, so the answer to Q1 is quoted as 286 s rather than 285.5 s. The error is dominated by a length that can be measured directly, not by the model, and that is the distinction the section exists to make: the physics is worth 6 %, the measurement is worth 10 %.

7.3 Limits of validity

Two limits bound the result. The quasi-steady assumption fails once the venting time approaches the acoustic transit time of the module, $L/c_s \approx 30\,\mathrm{ms}$; that is four orders of

Table 5: Error budget for $t_{0.3}$ at $d = 10\,\mathrm{cm}$. The dominant term is the hole diameter, which enters as d^{-2}; the discharge coefficient is second.

Source	Magnitude (%)	Effect on $t_{0.3}$ (%)	Handling
Hole diameter	5	10.5	geometric measurement
Discharge coefficient	6	6.3	calibrated, Sec. 7.1
Module volume	2	1.9	module geometry
Wall temperature	3	0.5	lumped model, Eq. (4)
Numeric	–	0.09	step halving
Total (quadrature)	–	12.4	dominant: d

magnitude below $t_{0.3}$ at 10 cm but only a factor of twenty at 1 m, so the model must not be pushed to metre-scale breaches without a compressible treatment. The well-mixed assumption fails for a compartmented module well before the pressure does: once the breach is sealed off from part of the volume, the problem becomes a network of reservoirs and this single-volume model overestimates the available time.

There is also a modelling limit worth stating plainly. Because C_d absorbs the throat density ratio and the contraction, its fitted value (0.82) is not comparable with the handbook value for a sharp-edged orifice (0.61): the two differ by the factor $(2/(\gamma+1))^{1/(\gamma-1)} = 0.63$ that the lumped form hides. A reader who substitutes the handbook number into (1) will get a time that is 34 % too long.

8 Strengths and Weaknesses

Strengths

(1) The model is built in three explicit layers, and each layer is priced: wall heat transfer moves $t_{0.3}$ by 2 % at 10 cm and 5 % at 5 cm, so a reader can see what was gained rather than take the complexity on trust.

(2) Every assumption carries a quantitative validity check (Table 1); none is merely asserted, and the one that is weakest (C_d constancy) is the one that is calibrated.

(3) The result is reduced to a scaling law and a single dimensionless time, $t_{0.3}/\tau = 1.41$, so the crew can evaluate a different module without running the code.

(4) The numerical error is two orders of magnitude below the physical error (0.09 % against 12 %), which is where it should be for a problem of this kind.

Weaknesses

(1) The wall heat transfer is first order with a single lumped time constant; with τ_w uncertain by 30 % the answer moves by under 1 % at both 10 cm and 5 cm, so the thermal layer earns its place through the trend in the exponent rather than through the nominal number.

(2) The lumped C_d absorbs the throat density ratio and the contraction, so its fitted value

cannot be compared with handbook data; the model is therefore calibrated for a sharp-edged circular hole and would need a new fit for a torn or irregular breach.

(3) The single-volume assumption fails for a compartmented module, so the model must not be applied there without a network extension, and it says nothing about the direction of the jet or the forces on the structure once the flow is established.

9 Conclusion

(1) A 10 cm breach in a 628 m^3 module at 101 kPa drops the internal pressure to $0.3\,p_0$ in 286 s, and the time scales as $t_{0.3} \propto d^{-1.97}$ between 5 cm and 20 cm.

(2) A crew needing 120 s to seal the breach keeps a margin of 166 s at 10 cm; the margin vanishes at 15.4 cm, above which the module must be abandoned.

(3) The answer is uncertain at the $\pm 12\,\%$ level, with 10 % of that coming from the hole diameter and 6 % from the discharge coefficient; measuring the hole is worth more than refining the model.

Because the two governing groups are γ and $\Pi_T = \tau/\tau_w$, the same numbers transfer to any rigid volume venting into vacuum once its size and hole diameter are expressed through τ — the result is a design rule for breach response, not a single-case calculation.

References

[1] Ascher H. Shapiro. *The Dynamics and Thermodynamics of Compressible Fluid Flow*, volume 1. Ronald Press, New York, 1953.

[2] John D. Anderson. *Modern Compressible Flow: With Historical Perspective*. McGraw-Hill, New York, 3 edition, 2003.

[3] Lev D. Landau and Evgeny M. Lifshitz. *Fluid Mechanics*, volume 6 of *Course of Theoretical Physics*. Pergamon Press, Oxford, 2 edition, 1987.

A Derivation of the orifice speed

Starting from the steady energy equation for an adiabatic nozzle with no work, $h + u^2/2 = h_0$, and using $h = c_p T$ with $T/T_0 = (p/p_0)^{(\gamma-1)/\gamma}$, the exit speed is

$$u^2 = \frac{2\gamma}{\gamma - 1}\frac{p_0}{\rho_0}\left[1 - \left(\frac{p_a}{p}\right)^{(\gamma-1)/\gamma}\right], \tag{7}$$

which is Eq. (2) written at the reservoir state. The bracket is largest when the flow chokes, at $p/p_a = ((\gamma+1)/2)^{\gamma/(\gamma-1)}$; substituting that ratio gives

$$u_{\mathrm{ch}}^2 = \frac{2\gamma}{\gamma+1}\frac{p_0}{\rho_0} = \frac{2\gamma}{\gamma+1} R T_0, \tag{8}$$

that is, Eq. (3), whose speed no longer depends on the ambient pressure. Differentiating the mass balance $\mathrm{d}m/\mathrm{d}t = -\rho_e A u$ with $m = \rho V$ and using $\rho \propto p^{1/\gamma}$ gives Eq. (1), with the ratio $\rho_e/\rho = (2/(\gamma+1))^{1/(\gamma-1)} = 0.63$ absorbed into C_d — which is why the fitted coefficient is not the handbook one.

B Numerical code

The integration quoted in Section 5 is performed by the script below; the listing is included with `lstinputlisting` so the printed code cannot drift from the version that produced the numbers.

```
"""Quasi-steady blow-down of a rigid, well-mixed module through a sharp orifice.

The model is Eqs. (1)-(3) of the paper: mass balance for the reservoir,
choked and subsonic orifice speed, and a first-order wall heat transfer.
Integrated with explicit RK4 and a fixed step small enough that halving it
changes t_0.3 by less than 0.1 % (see Section 5).

Run: python3 blowdown.py # nominal case, prints t_0.3
"""

import math

P_ATM = 101325.0
GAMMA = 1.4
R_AIR = 287.05


def exit_speed(p, p_a, T):
    """Exit speed (m/s) from the reservoir state, choked or subsonic."""
    ratio = p_a / p if p > 0 else 1.0
    critical = (2.0 / (GAMMA + 1.0)) ** (GAMMA / (GAMMA - 1.0))
    if ratio <= critical:
        return math.sqrt(GAMMA * R_AIR * T) * math.sqrt(
            2.0 / (GAMMA + 1.0))
    return math.sqrt(2.0 * GAMMA / (GAMMA - 1.0) * R_AIR * T
                     * (1.0 - ratio ** ((GAMMA - 1.0) / GAMMA)))


def rhs(p, T, A, V, Cd, p_a, T_w, tau_w, dt):
    """dp/dt and dT/dt for the reservoir."""
    u = exit_speed(p, p_a, T)
    dp = -GAMMA * p * A * Cd / V * u
    dT = (GAMMA - 1.0) / GAMMA * T / p * dp - (T - T_w) / tau_w
    return dp, dT


def blowdown(p0=P_ATM, V=628.0, d=0.10, Cd=0.82, T0=293.0, T_w=293.0,
             tau_w=300.0, dt=0.5):
    """Return (t, p, T) arrays until p falls to 0.3 p0."""
```

```
    A = math.pi * d ** 2 / 4.0
    p, T, t = p0, T0, 0.0
    times, pressures, temps = [t], [p], [T]
    while p > 0.3 * p0:
        k1p, k1T = rhs(p, T, A, V, Cd, 0.0, T_w, tau_w, dt)
        k2p, k2T = rhs(p + dt / 2 * k1p, T + dt / 2 * k1T, A, V, Cd, 0.0,
                       T_w, tau_w, dt)
        k3p, k3T = rhs(p + dt / 2 * k2p, T + dt / 2 * k2T, A, V, Cd, 0.0,
                       T_w, tau_w, dt)
        k4p, k4T = rhs(p + dt * k3p, T + dt * k3T, A, V, Cd, 0.0, T_w,
                       tau_w, dt)
        p += dt / 6 * (k1p + 2 * k2p + 2 * k3p + k4p)
        T += dt / 6 * (k1T + 2 * k2T + 2 * k3T + k4T)
        t += dt
        times.append(t)
        pressures.append(p)
        temps.append(T)
    return times, pressures, temps


if __name__ == "__main__":
    t, p, T = blowdown()
    print(f"t_0.3 = {t[-1]:.1f} s for d = 10 cm, V = 628 m^3")
```

Listing 1: Blow-down integration used for all results in Section 6.

C Parameter sweep data

Table 6: The three diameters used in Table 4. The last column multiplies the measured time by $(d/10\,\mathrm{cm})^{1.97}$; it is constant to 1 %, which is what makes the power law a valid summary over this range.

d (cm)	$t_{0.3}$ (s)	$t_{0.3}(d/10\,\mathrm{cm})^{1.97}$ (s)
5	1108	283
10	286	286
20	73	284